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Decision letter (Initial Submission)

AlGaN/GaN Hall-Effect Sensor for In-Situ Magnetic Field Monitoring of the HSX Stellarator

Subject IEEE Sensors Letters: SENSL-26-07-RL-1061

Date sent 22 July 2026 at 18:13 GMT-7

From sensl-admin@ieee.org

To timzhao@stanford.edu

CC giacomo.langfelder@polimi.it

CC

23-Jul-2026

Dear Mr. Zhao:

"AlGaN/GaN Hall-Effect Sensor for In-Situ Magnetic Field Monitoring of the HSX Stellarator" (Manuscript ID SENSL-26-07-RL-1061), which you submitted to IEEE Sensors Letters, has been reviewed.

Unfortunately, we must decline the manuscript for publication at this time. However, because we believe the work has merit, we invite you to consider submitting a revised manuscript that takes the reviewers' comments into consideration. It would be given a new Manuscript ID and reviewed again.

After you have revised your manuscript, please submit it by going to the submission site.

<https://ieee.atyponrex.com/submission/submissionBoard/REX-PROD-2-B16111D9-5CC0-4019-A52D-6A06D1BF6EDB-4EF4E7AF-1262-48DD-8A61-4B1F02BADD1F-23716/current?idtype=external>

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Thank you for sending your manuscript to us for consideration. We look forward to further contributions from you in the future.

Sincerely,

Dr. Giacomo Langfelder
AEIC, IEEE Sensors Letters

Manuscript Title: AlGaIn/GaN Hall-Effect Sensor for In-Situ Magnetic Field Monitoring of the HSX Stellarator

ID: SENSL-26-07-RL-1061

Editor's comments to author, if any
Associate Editor

Comments to the Author:

Please look through the reviewer comments and address all concerns. The main concern I share with the reviewers is novelty of this research. This paper is only showing changes in sensing values, but not in terms of what it is intended to sense (magnetic field). This could be good as a conference paper scope, but IEEE Sensors Letters requires a fully finished study of the intended sensing output. To address one of the reviewers, I would highly recommend to better communicate how this compares to other GaN sensors. A comparison table comparing performance would be highly useful here and also help with novelty. I have a few follow-on questions: what prevented the team from doing repetitive tests of the sensor to get statistical data on its performance (only one module was tested) - without repetition, we cannot have confidence in repeatability of its performance across fabrication iterations (if this is established in previous literature, it needs to be clearly shown in the paper on its own here) and why is there not a bench top calibration of the probe at the minimum. While the sensor is showing different magnetic field dynamics with the plasma, that is interesting from a plasma physics perspective, but the focus in a 4 page letters for sensors should be on the sensor itself. I believe Figure 5 is useful for the general audience to understand more about the various magnetic field environments occurring in HSX, but it does need to show it in magnetic field values (even if uncertainty regions are included, it will help the readers understand the order of magnitude they are seeing here). In general, these various changes come down to sensor bandwidth, parasitics, etc. that influence how well it can track magnetic field changes and how that compares to previous sensors of this class. Your packaging could potentially be novel if additional care compared to previous work improves parasitics in terms of wire bonding, bond pad topology, etc. It's a full sensor system you have here. For repeatability, I understand each shot is unique in HSX and you cannot perfectly repeat three iterations of the sensor in the same spot or even in parallel due to most likely space limitations. However, bench top testing of three fabrication iterations (unless only one sensor was made - please explain why this would be the case if it is) and

showing how close responses are to each other under the same controlled field would help build confidence that this sensor is repeatable.

Reviewers' comments to author, if any (please also check your Author Center for other files):

Reviewer: 1

Comments to the Author

General comments: The paper is very well written and clear about what it is conveying. The device itself is novel and unique to my knowledge. A magnetic field probe that can accurately measure low frequency fields in harsh plasma environments would be a great addition to the field. I do feel that the work is missing some basic characterization analysis. If this can be provided I think the paper would be significantly improved.

Key Point:

Was there any attempt to perform a benchtop calibration of this probe? I would consider a benchtop calibration, even if it can only be done for a small frequency range or even just DC, to be the first step in the development of a sensor. While voltage outputs are presented, there is no data regarding what the magnetic field strength actually is. I am not requiring that this data be obtained and added to the paper but if it already exists (or if you are inclined to obtain it), I think the quality of the paper would be significantly enhanced. Given the page limit, I think that discussion of the results of a benchtop calibration should suffice (no figures should be necessary unless S_v has a significant frequency dependence).

In lieu of a data from a benchtop calibration, the hall sensor's measurement could be compared to the magnetic field measurement of a conventional field probe in the HSX device. Is there data from a B-dot probe or equivalent that the hall probe could be compared to? This could at least give some bounds on the field being measured by the hall probe? The comparison to stored energy shows temporal consistency in signal strength changes but there is no where in the paper a direct magnetic field measurement comparison.

In summation, it would just be good to see some discussion or analysis of the device's response compared to a traditional magnetic field measurement, a 1:1 comparison. To reiterate, I think the work is still worth publishing if this is not added but the paper's quality would be significantly enhanced if it does get added.

Minor points:

1: Please call out the Endler W7x reference again when you mention Mirnov coils. This is a good reference to explain what Mirnov coils are and since they are presented as a comparison for your hall sensor, it would be better if the reader did not have to hunt down the ideal reference.

2: How was the 1 MHz bandwidth established? The amplifiers you listed have frequency capabilities well above 1 MHz. Is the limitation based on the device itself? Please discuss how you determined this limit. If this information is already in one of the referenced papers, it is sufficient to explicitly list that reference when you state the bandwidth limit.

Stylistic note:

The plots in fig 5 could be combined into 3 overlaid plots that use both left and right y-axes. It might make visual comparison easier. Not a requirement, just a thought.

I think the device you've developed and the work you've done here will be a significant contribution to the field. Thank you for your hard work and effort to advance these capabilities.

Reviewer: 2

Comments to the Author

This paper mainly uses 2DEG based Hall devices to detect dynamic pulsed magnetic fields. The

development and performance of GaN semiconductor Hall magnetic sensors have been reported previously, and it has been proven that they can detect rapidly changing magnetic fields. Therefore, this paper lacks sufficient novelty. In addition, the experimental detection data in the paper is insufficient, lacking in repetitive testing results and accuracy calibration process. Finally, there is insufficient literature research in the paper, and many references to GaN Hall devices are not cited. So this version should be rejected.

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